Data-Backed Journal-Style Replication of *Effects of Search Timing on Innovation*

Codex journal-replication workflow

2026-04-04

Abstract

This manuscript expands the local replication folder for Katila and Chen (2008) into a long journal-style paper using only processed datasets already present in the workspace. The current local batch contains 4,839 observations across 167 entities, and the batch headline estimate is -0.0854 for ahead vs sync on event quality with p = 0.1178. The local event-level analogue produces weaker timing effects than the source paper's argument would suggest, especially for event quality. The draft imitates the structure of the source article while documenting the exact places where local replication decisions depart from direct source facts.

Keywords: search timing; rivals; ahead versus behind; event-level innovation; local calculated replication

1 Paper-First Reading Protocol

The workflow for this manuscript is intentionally ordered. The source paper is read first, the required empirical objects are identified second, the local files are verified third, and only then is the replication paper written from the checked evidence chain.

Step	Action	Evidence
1	Read the original paper before touching the local estimates.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/katilachenasq2008.pdf
2	Extract the source paper's theoretical logic, sample definition, variable definitions, and estimand.	Effects of Search Timing on Innovation: The Value of Not Being in Sync with Rivals
3	Match each required empirical object to a local executable file and verify that the file exists.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_analysis_dataset.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_model_coefficients.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_sample_summary.csv

Step	Action	Evidence
4	Write the replication paper only from the checked local files and calculated outputs.	The manuscript, markdown notes, and PDF are generated from the verified file set.

Ordered protocol used for this replication paper.

2 Required Data and Local File Manifest

This section answers the basic replication question of where the data are. Each empirical object is mapped to a concrete local file before the estimates are interpreted. Workspace root: C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions

Stage	NeededFor	FilePath	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	patent literature/katilachenasq2008.pdf	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_analysis_dataset.csv	Yes	Executable local dataset

Stage	NeededFor	FilePath	Exists	Status
Coefficient export	Recover the reported local coefficients and p-values	patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_model_coefficients.csv	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_sample_summary.csv	Yes	Executable local result file

Exact local files used by the manuscript.

3 Introduction

The source paper asks whether moving ahead of, in sync with, or behind rivals changes innovative output and quality. The point of the present draft is not to pretend that every original source file has been recovered. It is to turn the executable local evidence into a paper that preserves theory, data architecture, variable mapping, and empirical claims in one place. The current local batch contains 4,839 observations across 167 entities, and the batch headline estimate is -0.0854 for ahead vs sync on event quality with $p = 0.1178$. The local event-level analogue produces weaker timing effects than the source paper's argument would suggest, especially for event quality.

4 Research Positioning and Contribution

The replication matters because timing relative to rivals is a relational construct that is easy to cite and hard to rebuild transparently from local event data. The paper's contribution is an explicit event-level reconstruction of ahead, sync, and behind timing that shows what the timing mechanism looks like in a local patent-based analogue. The writing stance follows a conservative paper-first workflow: the source paper defines the theoretical target, the local files define the executable boundary, and the replication paper names the distance between the two rather than smoothing it over.

5 Original Paper, Theory, and Published Estimand

The focal source article is [1]. The source paper is Effects of Search Timing on Innovation: The Value of Not Being in Sync with Rivals by Riitta Katila and Eric L. Chen in Administrative Science Quarterly . Timing relative to rivals is the substantive core of the original design. The original paper studies both innovation quantity and innovation quality. Ahead, sync, and behind are relational timing categories. A useful replication discipline is to keep source facts and replication decisions separate. Source facts describe what the paper says. Replication decisions describe what the local executable workflow must do because it is limited to processed datasets already present in the workspace.

- Timing relative to rivals is the substantive core of the original design.
- The original paper studies both innovation quantity and innovation quality.
- Ahead, sync, and behind are relational timing categories.

DirectSourceFact	LocalReplicationDecision
Timing relative to rivals is central.	The local event file reconstructs timing from recombination-combination first use.
Both quantity and quality are studied.	Event patent counts and mean inventive impact are the executable outcomes.
The categories are ahead, sync, and behind.	Events without rival comparators are excluded because timing categories cannot be assigned.

Direct source facts are kept separate from local replication decisions.

6 Decision Register

The table below makes the major design choices explicit before any coefficient is interpreted.

DecisionPoint	LocalSpecification
Research question	The source paper asks whether moving ahead of, in sync with, or behind rivals changes innovative output and quality.
Unit of analysis	Search event

DecisionPoint	LocalSpecification
Sample frame	First-use recombination events with rival-comparable timing categories in the local event panel.
Time window	1986 to 2016 in the local executable panel.
Outcome definition	Event patent count and event mean inventive impact.
Main explanatory design	Event-level models comparing ahead and behind timing categories to the sync benchmark.
Controls and fixed effects	Event breadth and field-experience controls, with available firm and year structure carried into estimation.
Exclusion or substitution logic	Events without a valid rival timing comparison are excluded because ahead, sync, and behind cannot be assigned honestly.

Decision register for the local replication design.

7 Local Data Inventory and Sample Reconstruction

The local analogue reconstructs first-use events for recombination combinations and classifies each event relative to rival timing. Every number in this manuscript is backed by local files in patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing or by the explicitly referenced processed regression panel.

- Event-level dataset built from first observed firm use of each recombination combination.
- Timing categories created by comparing event year against rival first-use years.
- Event outcomes measured by patent count and mean inventive impact.

File	Role
katila_chen_2008_analysis_dataset.csv	Event-level timing panel
katila_chen_2008_model_coefficients.csv	Quantity and quality coefficient table
katila_chen_2008_sample_summary.csv	Timing-group raw means

Local executable files used by this paper.

Timing	Events	MeanEventPatents	MeanEventImpact
sync	444.0000	1.1261	4.9553
behind	3034.0000	1.0906	4.4235
ahead	1361.0000	1.0705	4.5928

Sample reconstruction table from the local results folder.

8 Variable Mapping, Derivations, and Replication Decisions

The variable map is the bridge between the source article and the executable local workflow.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Search timing	Ahead / sync / behind relative to rivals	timing	Compare each firm's first-use year for a combination to rivals' first-use years.
Innovation quantity	Event patent count	event_patents	Count patents inside each event.
Innovation quality	Event mean inventive impact	event_mean_impact	Average patent-level impact inside each event.
Search breadth	Event breadth	event_breadth	Carry forward the local event breadth measure.
Field experience	Event field experience	event_field_exp	Carry forward the local event field-experience measure.

Variable mapping from source constructs to local measures. Event-level models comparing ahead and behind timing categories with the sync benchmark while controlling for event breadth and field experience.

$$Y_e = \beta_1 \mathbb{1}[\text{Ahead}_e] + \beta_2 \mathbb{1}[\text{Behind}_e] + \beta_3 \text{EventBreadth}_e + \beta_4 \text{EventFieldExp}_e + \alpha_f + \gamma_t + \varepsilon_e. \quad (1)$$

9 Empirical Design and Identification Limits

The local design remains strongest when it preserves the original unit of analysis and interaction logic, even if some measurement layers are reconstructed with defensible analogues rather than identical source files.

- The local event environment is a patent-recombination analogue rather than the source paper's original innovation-search setting.
- Some combinations have sparse rival overlap, which limits timing variation.
- The estimates are best read as a calculated analogue of the timing mechanism.

10 Main Results

Ahead versus sync is estimated at -0.0291 with p = 0.1416 in model katila_chen_event_patents. Behind versus sync is estimated at -0.0490 with p = 0.0114 in model katila_chen_event_patents. Event breadth is estimated at 0.0079 with p = 0.2042 in model katila_chen_event_patents. Event field experience is estimated at 0.0360 with p = ¡0.001 in model katila_chen_event_patents. Ahead versus sync is estimated at -0.0854 with p = 0.1178 in model katila_chen_event_quality. Behind versus sync is estimated at -0.0704 with p = 0.1208 in model katila_chen_event_quality. Event breadth is estimated at 0.1289 with p = ¡0.001 in model katila_chen_event_quality. Event field experience is estimated at 0.0147 with p = 0.2721 in model katila_chen_event_quality.

model	Term	estimate	std_error	p_value
katila_chen_event_patents	Ahead versus sync	-0.0291	0.0198	0.1416
katila_chen_event_patents	Behind versus sync	-0.0490	0.0194	0.0114
katila_chen_event_patents	Event breadth	0.0079	0.0062	0.2042
katila_chen_event_patents	Event field experience	0.0360	0.0063	0.0000
katila_chen_event_quality	Ahead versus sync	-0.0854	0.0530	0.1178
katila_chen_event_quality	Behind versus sync	-0.0704	0.0441	0.1208

model	Term	estimate	std_error	p_value
katila_chen_event_quality	Event breadth	0.1289	0.0195	0.0000
katila_chen_event_quality	Event field experience	0.0147	0.0132	0.2721

Key coefficient table used in the main text.

Term	Model	Estimate	PValue	Signal	Reading
Ahead versus sync	katila_chen_event_patents	-0.0291	0.1416	Weak	Negative estimate with p = 0.1416.
Behind versus sync	katila_chen_event_patents	-0.0490	0.0114	Moderate	Negative estimate with p = 0.0114.
Event breadth	katila_chen_event_patents	0.0079	0.2042	Weak	Positive estimate with p = 0.2042.
Event field experience	katila_chen_event_patents	0.0360	0.0000	Strong	Positive estimate with p = ¡0.001.
Ahead versus sync	katila_chen_event_quality	-0.0854	0.1178	Weak	Negative estimate with p = 0.1178.

Term	Model	Estimate	PValue	Signal	Reading
Behind versus sync	katila_chen_event_quality	-0.0704	0.1208	Weak	Negative estimate with p = 0.1208.
Event breadth	katila_chen_event_quality	0.1289	0.0000	Strong	Positive estimate with p = ¡0.001.
Event field experience	katila_chen_event_quality	0.0147	0.2721	Weak	Positive estimate with p = 0.2721.

Interpretive summary of the main coefficients.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Event patents	4839.0000	1.0882	0.4049	1.0000	1.0000	1.0000	1.0000	13.00
Event total impact	4839.0000	5.0087	8.2191	1.0000	3.0000	6.0000	0.0000	200.00
Event mean impact	4839.0000	4.5199	6.8945	1.0000	3.0000	6.0000	0.0000	200.00
Event breadth	4839.0000	2.4650	0.7319	2.0000	2.0000	3.0000	2.0000	10.00
Event field experience	4839.0000	2.7288	1.4265	1.7918	2.7081	3.6889	0.0000	7.86

Descriptive statistics for the main variables.

Quantile	N	MeanRegressor	MeanOutcome
1.0000	4411.0000	2.2909	4.4410
2.0000	428.0000	4.2593	5.3328

Quantile summary linking the main regressor to the local outcome.

11 Claim-Evidence Audit

This audit prevents the paper from blurring together original-paper statements, local estimates, and replication judgments.

Claim	EvidenceType	EvidenceFile	AuditNote
The source paper studies: The source paper asks whether moving ahead of, in sync with, or behind rivals changes innovative output and quality.	Direct source fact	patent literature/katilachenasq2008.pdf	Recovered from the source article before local writing begins.
The local executable panel contains 4,839 observations across 167 entities.	Local calculated fact	patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_analysis_dataset.csv — patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_sample_summary.csv	Backed by the verified analysis panel and sample audit.

Claim	EvidenceType	EvidenceFile	AuditNote
The local headline estimate is -0.0854 for ahead vs sync on event quality with p = 0.1178.	Local calculated fact	patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_model_coefficients.csv	Backed by the exported coefficient file used in the paper.
The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable.	Replication decision	Source-vs-decision table and limitations section	Used only when the original pipeline cannot be matched byte-for-byte from current local files.

Claim-evidence audit for the current manuscript.

12 Comparison to the Published Paper and Remaining Gaps

The current local event panel does not yield strong evidence that being ahead of rivals improves event quality. The ahead coefficient is negative and weak in the quality model, while being behind is more clearly negative for event patent counts. The honest replication standard is to describe the gap without smoothing it over. A useful local paper tells the next researcher exactly which parts of the original design are recoverable now and which parts remain approximated.

13 Discussion and Conclusion

The manuscript turns a replication folder into a paper that can be uploaded, cited internally, and rerun later without guessing how the evidence was assembled. For Katila and Chen (2008) the bottom line is: The local event-level analogue produces weaker timing effects than the source

paper's argument would suggest, especially for event quality.

A Appendix A: Source Inventory

Stage	NeededFor	FilePath	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	patent literature/katilachenasq2008.pdf	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_analysis_dataset.csv	Yes	Executable local dataset
Coefficient export	Recover the reported local coefficients and p-values	patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_model_coefficients.csv	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_sample_summary.csv	Yes	Executable local result file

Source and local file manifest for the manuscript.

B Appendix B: Reading Protocol

Step	Action	Evidence
1	Read the original paper before touching the local estimates.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/katilachenasq2008.pdf
2	Extract the source paper's theoretical logic, sample definition, variable definitions, and estimand.	Effects of Search Timing on Innovation: The Value of Not Being in Sync with Rivals
3	Match each required empirical object to a local executable file and verify that the file exists.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_analysis_dataset.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_model_coefficients.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_sample_summary.csv

Step	Action	Evidence
4	Write the replication paper only from the checked local files and calculated outputs.	The manuscript, markdown notes, and PDF are generated from the verified file set.

Paper-first protocol followed before writing the manuscript.

C Appendix C: Decision Register

DecisionPoint	LocalSpecification
Research question	The source paper asks whether moving ahead of, in sync with, or behind rivals changes innovative output and quality.
Unit of analysis	Search event
Sample frame	First-use recombination events with rival-comparable timing categories in the local event panel.
Time window	1986 to 2016 in the local executable panel.
Outcome definition	Event patent count and event mean inventive impact.
Main explanatory design	Event-level models comparing ahead and behind timing categories to the sync benchmark.
Controls and fixed effects	Event breadth and field-experience controls, with available firm and year structure carried into estimation.

DecisionPoint	LocalSpecification
Exclusion or substitution logic	Events without a valid rival timing comparison are excluded because ahead, sync, and behind cannot be assigned honestly.

Expanded decision register for the manuscript.

D Appendix D: Sample Reconstruction

Timing	Events	MeanEventPatents	MeanEventImpact
sync	444.0000	1.1261	4.9553
behind	3034.0000	1.0906	4.4235
ahead	1361.0000	1.0705	4.5928

Detailed sample reconstruction table.

E Appendix E: Variable Mapping

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Search timing	Ahead / sync / behind relative to rivals	timing	Compare each firm's first-use year for a combination to rivals' first-use years.
Innovation quantity	Event patent count	event_patents	Count patents inside each event.
Innovation quality	Event mean inventive impact	event_mean_impact	Average patent-level impact inside each event.
Search breadth	Event breadth	event_breadth	Carry forward the local event breadth measure.
Field experience	Event field experience	event_field_exp	Carry forward the local event field-experience measure.

Detailed construct mapping.

F Appendix F: Descriptive Statistics

Variable	N	Mean	SD	P25	Median	P75	Min	M
Event patents	4839.0000	1.0882	0.4049	1.0000	1.0000	1.0000	1.0000	13.00
Event total impact	4839.0000	5.0087	8.2191	1.0000	3.0000	6.0000	0.0000	200.00
Event mean impact	4839.0000	4.5199	6.8945	1.0000	3.0000	6.0000	0.0000	200.00
Event breadth	4839.0000	2.4650	0.7319	2.0000	2.0000	3.0000	2.0000	10.00
Event field experience	4839.0000	2.7288	1.4265	1.7918	2.7081	3.6889	0.0000	7.86

Descriptive statistics appendix.

G Appendix G: Correlation Matrix

Variable	Event patents	Event total impact	Event mean impact	Event breadth	Event field experience
Event patents	1.000	0.313	0.032	-0.025	0.089
Event total impact	0.313	1.000	0.915	0.065	0.002
Event mean impact	0.032	0.915	1.000	0.074	-0.027
Event breadth	-0.025	0.065	0.074	1.000	-0.242
Event field experience	0.089	0.002	-0.027	-0.242	1.000

Correlation matrix for the main numeric variables.

H Appendix H: Yearly Audit

Period	V1	V2	V3	V4	V5
1986.0000	274.0000	1.1168	4.1745	2.3431	1.7241
1987.0000	313.0000	1.0767	3.8496	2.3642	2.0088
1988.0000	354.0000	1.1073	4.2290	2.2938	2.2682
1989.0000	411.0000	1.1168	3.8925	2.3382	2.3748
1990.0000	351.0000	1.1111	4.4128	2.3875	2.4820
1991.0000	351.0000	1.0855	4.6899	2.4444	2.6314
1992.0000	318.0000	1.0755	5.4591	2.4686	2.5444
1993.0000	321.0000	1.0903	5.5696	2.4984	2.7435
1994.0000	304.0000	1.0526	5.4940	2.5164	2.7160
1995.0000	242.0000	1.0826	6.3399	2.4917	2.8292
1996.0000	126.0000	1.0397	5.7262	2.5635	2.8326
1997.0000	111.0000	1.0811	5.5676	2.4685	3.0879
1998.0000	58.0000	1.0862	5.5891	2.4310	2.7553
1999.0000	138.0000	1.0725	5.7114	2.5652	2.8623
2000.0000	119.0000	1.0840	6.4762	2.4958	2.8887
2001.0000	94.0000	1.0319	4.0426	2.3936	3.0980
2002.0000	92.0000	1.0652	4.1467	2.7065	2.9604
2003.0000	82.0000	1.0122	4.4024	2.4756	3.3064
2004.0000	37.0000	1.0000	3.0541	2.4595	3.1022
2005.0000	43.0000	1.3256	5.6592	2.5116	3.0457
2006.0000	36.0000	1.0556	3.0694	2.5278	3.5952
2007.0000	38.0000	1.0000	2.7368	2.5000	3.3992
2008.0000	79.0000	1.2025	2.2443	2.6076	3.4918
2009.0000	24.0000	1.0417	2.5000	2.9167	3.5316
2010.0000	54.0000	1.0370	1.8889	2.9074	3.4028
2011.0000	75.0000	1.0400	4.3467	2.6133	4.0232

Period	V1	V2	V3	V4	V5
2012.0000	101.0000	1.0891	2.8383	2.6139	4.0222
2013.0000	78.0000	1.1026	2.4679	2.6795	3.5961
2014.0000	69.0000	1.1449	2.5352	2.8261	4.0339
2015.0000	100.0000	1.0300	1.9700	2.5400	4.0590
2016.0000	46.0000	1.2174	1.5362	2.6087	3.8081

Year-by-year sample audit.

I Appendix I: Grouped Outcome Patterns

Quantile	N	MeanRegressor	MeanOutcome
1.0000	4411.0000	2.2909	4.4410
2.0000	428.0000	4.2593	5.3328

Quantile summary for the main regressor.

Timing	Events	MeanEventPatents	MeanEventImpact
sync	444.0000	1.1261	4.9553
behind	3034.0000	1.0906	4.4235
ahead	1361.0000	1.0705	4.5928

Raw group means from the local summary file.

J Appendix J: Full Coefficient Table

term	estimate	std_error	statistic	p_value	model
timing::ahead	-0.0291	0.0198	-1.4698	0.1416	katila_chen_event_pate
timing::behind	-0.0490	0.0194	-2.5304	0.0114	katila_chen_event_pate
event_breadth	0.0079	0.0062	1.2696	0.2042	katila_chen_event_pate
event_field_exp	0.0360	0.0063	5.7486	0.0000	katila_chen_event_pate
timing::ahead	-0.0854	0.0530	-1.6102	0.1178	katila_chen_event_qual
timing::behind	-0.0704	0.0441	-1.5969	0.1208	katila_chen_event_qual
event_breadth	0.1289	0.0195	6.6249	0.0000	katila_chen_event_qual
event_field_exp	0.0147	0.0132	1.1188	0.2721	katila_chen_event_qual

Full coefficient dump exported from the local estimation script.

K Appendix K: Missingness Audit

Variable	Missing	MissingShare
Event patents	0.0000	0.0000
Event total impact	0.0000	0.0000
Event mean impact	0.0000	0.0000
Event breadth	0.0000	0.0000
Event field experience	0.0000	0.0000

Missingness audit for the variables summarized in the manuscript.

L Appendix L: Coverage of the Most Observed Entities

Entity	Observations	FirstPeriod	LastPeriod
6066.0000	393.0000	1986.0000	2016.0000
5047.0000	275.0000	1986.0000	2015.0000
7585.0000	202.0000	1986.0000	2012.0000
10499.0000	200.0000	1986.0000	2016.0000
4087.0000	182.0000	1986.0000	2013.0000
5073.0000	168.0000	1986.0000	2007.0000
4194.0000	151.0000	1986.0000	2010.0000
1300.0000	126.0000	1986.0000	2015.0000
1581.0000	117.0000	1986.0000	2004.0000
4060.0000	117.0000	1986.0000	2016.0000
4839.0000	113.0000	1986.0000	2016.0000
5606.0000	89.0000	1986.0000	2000.0000
4503.0000	87.0000	1986.0000	2013.0000
10983.0000	82.0000	1986.0000	2015.0000
11436.0000	82.0000	1986.0000	1997.0000
7475.0000	81.0000	1986.0000	1995.0000
7435.0000	76.0000	1986.0000	2015.0000
12141.0000	72.0000	1995.0000	2016.0000
1690.0000	70.0000	1990.0000	2016.0000
2285.0000	64.0000	1986.0000	2015.0000

Top entities by local observation count.

M Appendix M: Claim-Evidence Audit

Claim	EvidenceType	EvidenceFile	AuditNote
The source paper studies: The source paper asks whether moving ahead of, in sync with, or behind rivals changes innovative output and quality.	Direct source fact	patent literature/katilachenasq2008.pdf	Recovered from the source article before local writing begins.
The local executable panel contains 4,839 observations across 167 entities.	Local calculated fact	patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_analysis_dataset.csv — patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_sample_summary.csv	Backed by the verified analysis panel and sample audit.
The local headline estimate is -0.0854 for ahead vs sync on event quality with p = 0.1178.	Local calculated fact	patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_model_coefficients.csv	Backed by the exported coefficient file used in the paper.

Claim	EvidenceType	EvidenceFile	AuditNote
The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable.	Replication decision	Source-vs-decision table and limitations section	Used only when the original pipeline cannot be matched byte-for-byte from current local files.

Expanded claim-evidence audit.

N Appendix N: Source Inventory Note

Source Inventory - paper: Katila and Chen (2008) - source title: Effects of Search Timing on Innovation: The Value of Not Being in Sync with Rivals - source file: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/katilachenasq2008.pdf‘ ## Paper-First Reading Protocol - Step 1: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/katilachenasq2008.pdf‘ — needed for: Read the original paper before touching the local estimates. — exists: n/a — status: workflow step - Step 2: ‘Effects of Search Timing on Innovation: The Value of Not Being in Sync with Rivals‘ — needed for: Extract the source paper’s theoretical logic, sample definition, variable definitions, and estimand. — exists: n/a — status: workflow step - Step 3: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_ana
— C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/20
04-04/katila-chen-2008-search-timing/results/katila_chen_2008_model_coefficients.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_sample_summary.csv‘ — needed for: Match each required empirical object to a local executable file and verify that the file exists. — exists: n/a — status: workflow step - Step 4: ‘The manuscript, markdown notes, and PDF are generated from the verified file set.‘ — needed for: Write the replication paper only from the checked local files and calculated outputs. — exists: n/a — status: workflow step ## Local files - Source article: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/katilachenasq2008.pdf‘ — needed for: Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices — exists: Yes — status: Direct source - Analysis panel: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026
04-04/katila-chen-2008-search-timing/results/katila_chen_2008_analysis_dataset.csv‘ — needed for: Estimate the local model and regenerate descriptive statistics — exists: Yes — status: Executable local dataset - Coefficient export: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_
— needed for: Recover the reported local coefficients and p-values — exists: Yes — status: Executable local result file - Sample summary: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate

Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_
— needed for: Audit sample construction and raw grouped moments — exists: Yes — status: Executable local result file

O Appendix O: Study Map Note

Study Map - research question: The source paper asks whether moving ahead of, in sync with, or behind rivals changes innovative output and quality. - positioning: The replication matters because timing relative to rivals is a relational construct that is easy to cite and hard to rebuild transparently from local event data. - contribution: The paper's contribution is an explicit event-level reconstruction of ahead, sync, and behind timing that shows what the timing mechanism looks like in a local patent-based analogue. - local design: The local analogue reconstructs first-use events for recombination combinations and classifies each event relative to rival timing. - estimand: Event-level models comparing ahead and behind timing categories with the sync benchmark while controlling for event breadth and field experience. - headline estimate: '-0.0854' with 'p = 0.1178'. ## Decision Audit - Research question: The source paper asks whether moving ahead of, in sync with, or behind rivals changes innovative output and quality. - Unit of analysis: Search event - Sample frame: First-use recombination events with rival-comparable timing categories in the local event panel. - Time window: 1986 to 2016 in the local executable panel. - Outcome definition: Event patent count and event mean inventive impact. - Main explanatory design: Event-level models comparing ahead and behind timing categories to the sync benchmark. - Controls and fixed effects: Event breadth and field-experience controls, with available firm and year structure carried into estimation. - Exclusion or substitution logic: Events without a valid rival timing comparison are excluded because ahead, sync, and behind cannot be assigned honestly. ## Source Versus Decision - direct source fact: Timing relative to rivals is central. - direct source fact: Both quantity and quality are studied. - direct source fact: The categories are ahead, sync, and behind. local replication decision: The local event file reconstructs timing from recombination-combination first use. local replication decision: Event patent counts and mean inventive impact are the executable outcomes. local replication decision: Events without rival comparators are excluded because timing categories cannot be assigned.

P Appendix P: Data Design Note

Data Design - Event-level dataset built from first observed firm use of each recombination combination. - Timing categories created by comparing event year against rival first-use years. - Event outcomes measured by patent count and mean inventive impact. - panel file used: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_analysis_dataset.csv‘ - coefficient file used: ‘C:/Users/bai/OneDr - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_model_coefficients.csv‘ - sample summary used: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/202 04-04/katila-chen-2008-search-timing/results/katila_chen_2008_sample_summary.csv‘ - observations: ‘4,839‘ - entities: ‘167‘ ## Required Data And Local File Mapping - Source article: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/katilachenasq2008.pdf‘ — needed for: Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices — exists: Yes — status: Direct source - Analysis panel: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_analysis_dataset.csv‘ — needed for: Estimate the local model and regenerate descriptive statistics — exists: Yes — status: Executable local dataset - Coefficient export: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_model_coefficients.csv — needed for: Recover the reported local coefficients and p-values — exists: Yes — status: Executable local result file - Sample summary: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008 — needed for: Audit sample construction and raw grouped moments — exists: Yes — status: Executable local result file

Q Appendix Q: Derivations Note

Derivations And Estimands - The local event-level analogue produces weaker timing effects than the source paper's argument would suggest, especially for event quality. - Event-level models comparing ahead and behind timing categories with the sync benchmark while controlling for event breadth and field experience. - model equations are reproduced in ‘paper/main.tex‘. ## Variable Map - Search timing -¿ ‘timing‘ : Compare each firm's first-use year for a combination to rivals' first-use years. - Innovation quantity -¿ ‘event_patents‘ : Count patents inside each event. - Innovation quality -¿ ‘event_mean_impact‘ : Average patent-level impact inside each event. - Search breadth -¿ ‘event_breadth‘ : Carry forward the local event breadth measure. - Field experience -¿ ‘event_field_exp‘ : Carry forward the local event field-experience measure. ## Claim Evidence Audit - Direct source fact: The source paper studies: The source paper asks whether moving ahead of, in sync with, or behind rivals changes innovative output and quality. — evidence: patent literature/katilachenasq2008.pdf - Local calculated fact: The local executable panel contains 4,839 observations across 167 entities. — evidence: patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_analysis_dataset.csv — patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_sample_summary.csv - Local calculated fact: The local headline estimate is -0.0854 for ahead vs sync on event quality with p = 0.1178. — evidence: patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_model_coefficients.csv - Replication decision: The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable. — evidence: Source-vs-decision table and limitations section

R Appendix R: Replication Decisions Note

Replication Decisions - The local event environment is a patent-recombination analogue rather than the source paper's original innovation-search setting. - Some combinations have sparse rival overlap, which limits timing variation. - The estimates are best read as a calculated analogue of the timing mechanism. - direct source facts are separated from local replication decisions in the manuscript. - no coefficient or table is fabricated. ## Source Facts Versus Local Decisions - source: Timing relative to rivals is central. - source: Both quantity and quality are studied. - source: The categories are ahead, sync, and behind. decision: The local event file reconstructs timing from recombination-combination first use. decision: Event patent counts and mean inventive impact are the executable outcomes. decision: Events without rival comparators are excluded because timing categories cannot be assigned.

S Appendix S: Local Execution Status Note

Local Execution Status - batch date: ‘2026-04-04‘ - status: long-form journal-style manuscript generated from local result files - current paper: ‘katila-chen-2008-search-timing‘

T Appendix T: Cleaned Output Tables Note

Cleaned Output Tables - paper-first reading protocol - required-data and file-manifest table - decision register - sample reconstruction table - local source inventory table - source facts versus replication decisions table - claim-evidence audit table - variable mapping table - key coefficient table - full coefficient table - descriptive statistics - correlation matrix - yearly audit - missingness audit - top-entity coverage table - quantile summary

U Appendix U: Build Notes

Build Notes

- batch date: ‘2026-04-04‘ - manuscript style: long-form journal-style replication draft - evidence rule: only processed local datasets and actual local estimates are used - workflow rule: read the source paper first, then map required datasets to verified local files, then write the paper from those files - skill path guiding the structure: ‘C:/Users/bai/.codex/skills/write-replication-journal-github/SKILL.md‘

V Appendix V: Manuscript Metadata

Field	Value
Paper	Katila and Chen (2008)
Source title	Effects of Search Timing on Innovation: The Value of Not Being in Sync with Rivals
Authors	Riitta Katila and Eric L. Chen
Outlet	Administrative Science Quarterly
Batch date	2026-04-04
Rows	4,839
Entities	167
Headline term	ahead vs sync on event quality
Headline estimate	-0.0854
Headline p-value	0.1178
Panel path	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/katila-chen-2008-search-timing/results/katila_chen_2008_analysis_dataset.csv

Metadata block for GitHub publishing and future reruns.

W Appendix W: Replication Summary Note

Research question: The source paper asks whether moving ahead of, in sync with, or behind rivals changes innovative output and quality. Local design: The local analogue reconstructs first-use events for recombination combinations and classifies each event relative to rival timing. Estimand: Event-level models comparing ahead and behind timing categories with the sync benchmark while controlling for event breadth and field experience. Local result focus: The local event-level analogue produces weaker timing effects than the source paper's argument would suggest, especially for event quality. Comparison note: The current local event panel does not yield strong evidence that being ahead of rivals improves event quality. The ahead coefficient is negative and weak in the quality model, while being behind is more clearly negative for event patent counts.

References

[1] Riitta Katila and Eric L. Chen. Effects of search timing on innovation: The value of not being in sync with rivals. *Administrative Science Quarterly*, 2008.